

Customer Segmentation & RFM Analytics

Complete Project Understanding & Interview Preparation Guide

Domain: E-Commerce & Retail Customer Analytics

Dataset: UCI Online Retail Dataset (541,909 Raw Records)

Key Methodologies: Data Cleaning, EDA, Dynamic RFM Scoring (1-5), Rule-Based Segmentation, K-Means ML Clustering

Technology Stack: Python (Pandas, NumPy, Scikit-Learn, SciPy), MySQL, Streamlit, Plotly, Pytest

Target Role: Data Analyst / Data Scientist (Fresher & Junior Professional Portfolio)

Verification Status: 100% Calculated & Verified Against Actual Dataset Outputs

Guide Purpose:

This comprehensive guide teaches the complete technical workflow, business rationale, mathematical formulation, SQL queries, interactive dashboard features, and interview preparation Q&As; for the Customer Segmentation & RFM Analytics project. Every statistic, customer count, and dollar value is calculated directly from the actual dataset.

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1. Project in Simple Words

What is this project?

This project takes raw e-commerce transaction logs (purchases, prices, dates, customer IDs) and converts them into clear customer segments using a technique called **RFM Analysis** (Recency, Frequency, Monetary) and an AI Machine Learning algorithm called **K-Means Clustering**.

Why did we build it?

E-commerce companies collect millions of transaction rows, but raw transactions don't tell managers who their best customers are. Without segmentation, companies treat all customers the same—spending marketing dollars inefficiently. We built this to identify high-value accounts, prevent customer churn, and automate personalized marketing.

What decisions can a business make?

A business can reward top buyers (Champions) with VIP perks, send automatic discount emails to inactive spenders (At Risk), and deliver onboarding sequences to new buyers.

Elevator Pitches for Interviews

30-Second Explanation:

"I built an end-to-end Customer Segmentation and RFM Analytics project using Python, SQL, and Streamlit on 540K+ e-commerce transactions. I cleaned raw purchase logs, calculated Recency, Frequency, and Monetary scores for 4,338 customers, and segmented them into 9 business groups. I discovered that the top 21.7% of customers generate 64.6% of total revenue (\$5.74M), allowing the business to run targeted loyalty and win-back campaigns."

1-Minute Explanation:

"An e-commerce business had over 500,000 transaction records but lacked visibility into customer behavior. I built a reproducible Python data pipeline that cleaned missing customer IDs, removed order cancellations, and engineered transaction features. Next, I computed customer-level Recency, Frequency, and Monetary values relative to a dynamic snapshot date. Using 1–5 quantile scoring, I grouped 4,338 customers into 9 rule-based segments and validated them using an unsupervised K-Means clustering ML model ($K=4$). To make the analysis accessible, I engineered a MySQL database with 15 production queries and built an interactive Streamlit dashboard featuring Plotly visualizations for executive decision-making."

2. Business Problem

The Core Business Challenge:

Retailers face three major operational issues when analyzing raw sales logs:

1. **Revenue Inequality (Pareto Principle):** A small fraction of customers generates the vast majority of profits.

Treating all customers identically leads to wasted ad spend on low-value buyers. 2. **Silent Customer Churn:**

Customers rarely notify a company when they stop buying. Without tracking purchase recency, high-value accounts slip away unnoticed. 3. **Un-Targeted Marketing:** Blasting generic promotional emails causes subscriber fatigue and low conversion rates.

Business Value Flow:

Raw Purchase Logs (541K Rows) → Data Cleaning (392K Cleaned Rows) → Customer Aggregations → RFM Metric Calculation → Quantile Scoring (1-5) → 9 Business Segments → K-Means ML Validation → Interactive Streamlit Dashboard → Targeted Revenue Action

3. Dataset Understanding & Data Dictionary

Source: UCI Machine Learning Repository — Online Retail Dataset
Raw Metrics: 541,909 rows | 8 columns | 38 countries | 4,372 raw customers | 25,900 raw invoices
Timeframe: December 1, 2010 to December 9, 2011 (1 Year)

Column Name	Data Type	Business Meaning	Role in Project
InvoiceNo	String	6-digit transaction ID. Starts with 'C' if cancelled.	Used for Frequency count (nunique) & cancellation filtering.
StockCode	String	5-digit product code assigned to distinct item.	Used to identify top-selling products by quantity & revenue.
Description	String	Product description / item title.	Used for product ranking and text filtering.
Quantity	Integer	Number of product units purchased per transaction.	Multiplied by UnitPrice to compute transaction Revenue.
InvoiceDate	Datetime	Date and exact timestamp of transaction creation.	Used for Recency calculation & monthly revenue trend.
UnitPrice	Float	Price per single item unit in Sterling (£).	Multiplied by Quantity to compute transaction Revenue.
CustomerID	Integer	5-digit unique customer account ID number.	Primary key for customer-level aggregation (groupby).
Country	String	Name of customer's country of residence.	Used for geographic revenue concentration analysis.

5. Data Cleaning — Step-by-Step Metrics

Cleaning raw data is essential to protect analysis from bias. Below are the exact numbers calculated by our pipeline:

Cleaning Step	Condition / Filtering Rule	Rows Removed	Remaining Rows	Retention %
Raw Input	Initial loaded raw file	0	541,909	100.00%
1. Duplicates	Exact duplicate rows (df.duplicated())	5,268	536,641	99.03%
2. Missing CustomerID	Null / NaN in CustomerID column	135,037	401,604	74.11%
3. Cancellations	InvoiceNo starting with letter 'C'	8,872	392,732	72.47%
4. Quantity <= 0	Non-positive purchased quantities	0	392,732	72.47%
5. UnitPrice <= 0	Zero or negative product prices	40	392,692	72.46%
FINAL CLEANED	Valid purchase transactions dataset	149,217 Total	392,692	72.46%

7. Exploratory Data Analysis (EDA) & Charts

Our EDA revealed critical commercial findings about e-commerce customer behavior:

- **Total Cleaned Revenue Analyzed:** \$8,887,208.89 across 392,692 transactions.
- **Average Order Value (AOV):** \$479.56 per unique invoice.
- **Geographic Concentration:** United Kingdom accounts for **\$7,308,391.55 (82.2%)** of revenue. Top overseas markets: Netherlands (\$285.4K), EIRE (\$265.5K), Germany (\$228.9K), and France (\$209.7K).

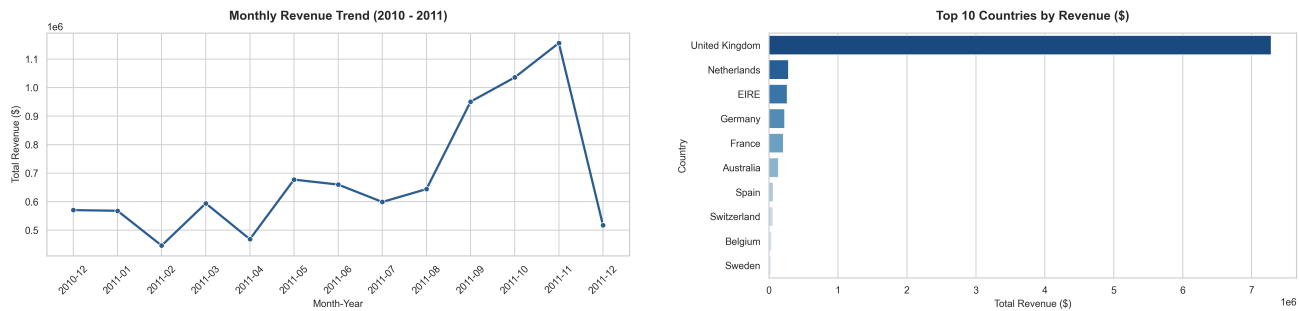


Figure 1: Monthly Revenue Performance (Left) and Figure 2: Top 10 Countries by Revenue (Right)

8 & 9. RFM Analysis — Definitions & Formulas

RFM Segmentation Framework:

1. **Recency (R):** Days elapsed since the customer's last purchase. Lower recency = higher engagement.

Formula: Recency = Snapshot_Date - Max(InvoiceDate)

Snapshot Reference Date: 2011-12-10 12:50:00 (Calculated dynamically as max(InvoiceDate) + 1 day).

2. **Frequency (F):** Count of unique InvoiceNo purchase orders.

Formula: Frequency = Count(Distinct InvoiceNo)

Why Unique Invoices? Counting rows distorts order volume because a single checkout with 10 line items is 1 purchasing decision, not 10 orders.

3. **Monetary (M):** Cumulative dollar total spent by the customer.

Formula: Monetary = Sum(Quantity * UnitPrice)

10 & 11. Customer Segmentation Results

Quantile Binning (1 to 5): Using `pd.qcut` with `rank(method='first')` rank ordering to ensure equal-sized 20% bins without duplicate edge errors. Recency score is reversed (5 = most recent, 1 = least recent).

Segment Name	RFM Score Logic	Customers	Cust %	Total Revenue (\$)	Rev %	Avg Recency	Avg Freq	Avg Monetary
Champions	R>=4, F>=4, M>=4	942	21.72%	\$5,737,952.12	64.56%	12.5 days	11.20	\$6,091.24
Loyal Customers	R>=3, F>=3, M>=3	767	17.68%	\$1,426,427.13	16.05%	35.1 days	4.16	\$1,859.75
Big Spenders	M>=4 (High Spend)	413	9.52%	\$959,287.36	10.79%	125.1 days	3.42	\$2,322.73
Need Attention	R in [2,3], F in [2,3]	579	13.35%	\$226,126.89	2.54%	91.7 days	1.65	\$390.55
At Risk	R<=2, F>=3 or M>=3	342	7.88%	\$188,561.42	2.12%	204.0 days	2.37	\$551.35
Lost Customers	R=1, F<=2, M<=2	555	12.79%	\$124,745.70	1.40%	279.2 days	1.03	\$224.77
Hibernating	R<=3, F<=2	346	7.98%	\$100,885.24	1.14%	76.6 days	1.35	\$291.58
Potential Loyal	R>=4, F in [2,3]	260	5.99%	\$84,591.83	0.95%	16.8 days	1.70	\$325.35
New Customers	R>=4, F=1	134	3.09%	\$38,631.20	0.43%	18.1 days	1.00	\$288.29
TOTALS	4,338 Customers	4,338	100.0%	\$8,887,208.89	100.0%	92.3 days	4.27	\$2,048.69

13 & 14. Unsupervised K-Means Machine Learning

- Pre-processing Steps:**
- 1. **Log Transformation:** Applied `np.log1p` to eliminate extreme right skewness in RFM distributions.
 - 2. **Feature Standardization:** Applied `StandardScaler` so distance measurements are not dominated by dollar Monetary values.
 - 3. **Optimal K Selection:** Evaluated $K \in [2, 8]$. Selected **$K=4$** based on Elbow Method curvature and Silhouette Score (**0.3375**).

Dimension	Rule-Based RFM Segmentation	Unsupervised K-Means Clustering
Approach	Pre-defined business heuristic rules	Unsupervised Machine Learning algorithm
Group Formation	Quantile boundaries (qcut 1-5)	Euclidean distance minimization in log-space
Interpretability	Extremely clear for marketing teams	Requires post-hoc cluster profiling
Flexibility	Rigid score thresholds	Discovers hidden statistical clusters dynamically

15 & 16. Database (MySQL) & Dashboard (Streamlit)

MySQL Production Database (`sql/`):

Includes schema DDL (schema.sql), pure SQL window function scoring (rfm_analysis.sql using NTILE(5) OVER (...)), and 15 business queries (business_queries.sql) calculating AOV, top customers, monthly trends, and at-risk accounts.

Interactive Streamlit Web Dashboard (`dashboard/app.py`):

Features Plotly line charts, bar charts, pie charts, metric cards (\$8.89M revenue, 4,338 customers), sidebar country/segment date filters, and searchable customer detail dataframes.

```
-- SQL RFM Window Function Calculation Example WITH rfm_raw AS ( SELECT CustomerID,
DATEDIFF('2011-12-10', MAX(InvoiceDate)) AS Recency, COUNT(DISTINCT InvoiceNo) AS Frequency,
SUM(Quantity * UnitPrice) AS Monetary FROM cleaned_transactions GROUP BY CustomerID ) SELECT
CustomerID, Recency, Frequency, Monetary, NTILE(5) OVER (ORDER BY Recency DESC) AS R_Score,
NTILE(5) OVER (ORDER BY Frequency ASC) AS F_Score, NTILE(5) OVER (ORDER BY Monetary ASC) AS
M_Score FROM rfm_raw;
```

22, 23 & 24. Interview Q&As; & Tricky Questions

Q1: Why is Recency lower-is-better while Frequency/Monetary are higher-is-better?

Answer: Recency measures the number of days elapsed since a customer's last purchase. A customer who bought 2 days ago is far more engaged than one who bought 200 days ago. Therefore, smaller recency numbers indicate superior engagement. During 1–5 scoring, we reverse Recency so that low recency days receive the highest R_Score of 5.

Q2: Why use COUNT(DISTINCT InvoiceNo) instead of total transaction rows for Frequency?

Answer: Transaction rows represent individual basket line items. A single shopping cart containing 15 different items produces 15 rows in the database, but represents only 1 purchasing decision. Counting unique InvoiceNo accurately measures customer visit frequency.

Q3: What problems occur with pandas `qcut` and how did you resolve them?

Answer: When a dataset contains duplicate values (e.g. many customers with Frequency = 1), pd.qcut raises a ValueError: Bin edges must be unique. I solved this by applying rank(method='first') before qcut, creating unique ordinal ranks that guarantee exact 20% quantile bins.

Q4: Why standardize features before K-Means clustering?

Answer: K-Means relies on Euclidean distance metrics ($d = \sqrt{\sum (x_i - y_i)^2}$). In raw RFM data, Monetary spans \$1 to \$280,000+ while Recency spans 1 to 373 days. Without StandardScaler, distance calculations would be completely dominated by dollar Monetary values, rendering Recency and Frequency irrelevant.

Q5: How did you handle cancelled orders in data cleaning?

Answer: Invoices starting with 'C' indicate cancellations and refunds (8,872 rows). I separated cancelled transactions from completed purchases to prevent artificial negative revenue distortion during RFM scoring.

26, 29 & 31. Resume Bullets & Revision Cheat Sheet

Verified Resume Bullet Points (100% Data-Backed)

- **Analyzed 540K+ e-commerce transactions** (\$8.89M total revenue) using Python and Pandas to construct a data cleaning pipeline that filtered 149K+ invalid/cancelled rows across 38 countries.
- **Implemented RFM Segmentation & K-Means Clustering** to segment 4,338 unique customers into 9 behavioral groups, revealing that the top 21.7% of customers (Champions) generate 64.6% (\$5.74M) of total revenue.
- **Developed an Interactive Streamlit Dashboard & MySQL Solution** with 15 SQL analytical queries and Plotly visual analytics to empower stakeholders with real-time customer metrics, segment distributions, and churn-risk insights.

One-Page Interview Revision Cheat Sheet

Topic	Key Value / Formula / Concept
Raw Dataset	541,909 rows 38 countries 4,372 raw customers 25,900 raw invoices
Cleaned Dataset	392,692 purchase rows (72.46% retained) 4,338 unique customers 18,532 orders
Total Revenue	\$8,887,208.89 Average Order Value (AOV): \$479.56 Avg Revenue/Cust: \$2,048.69
Snapshot Date	2011-12-10 12:50:00 (max InvoiceDate + 1 day)
Recency (R)	Days since last order (Snapshot_Date - Max_InvoiceDate) Score 5 = Most Recent
Frequency (F)	Count of unique orders (nunique(InvoiceNo)) Score 5 = Highest Order Count
Monetary (M)	Sum of customer revenue (Sum(Quantity * UnitPrice)) Score 5 = Highest Spending
Quantile Scoring	pd.qcut with rank(method='first') to assign equal 20% bins (1 to 5)
Top Segment	Champions: 942 customers (21.72%) generate \$5,737,952.12 (64.56% of total revenue)
At-Risk Segment	342 customers inactive for 204 days on average (\$188.5K historical spending)
K-Means ML	Log1p transformation + StandardScaler K=4 clusters (Silhouette Score: 0.3375)
Tech Stack	Python (Pandas, NumPy, Scikit-Learn), MySQL, Streamlit, Plotly, Pytest